Heterogeneous Furniture Transportation

Updated Problem Model, Generator and Genetic Algorithm

1. Interpretation of the updated requirements

The supplied vehicles differ in payload, load volume and fuel consumption. The suggested transport simulation includes depot returns and subsequent loads. The appropriate research direction is therefore a **heterogeneous-fleet multi-trip vehicle routing problem**, with furniture-specific capacity constraints. This replaces the previous identical-vehicle demonstration as the working model.

Vehicle	Length	Volume	Payload	Fuel L/100km
VW Caddy Cargo SWB	4.50 m	3.1 m3	680 kg	4.5-6.2
Mercedes Sprinter 315 CDI L2H2	5.93 m	10.5 m3	1,231 kg	9.0-9.1
MAN TGL 12.190 box	9.20 m	42.7 m3	5,480 kg	18-24

These are adopted instructor-provided parameters, not an independent verification of vehicle documents. Overall vehicle length is not usable cargo-bay length. Top speeds are recorded in the input but are not used as operating speeds.

Initial simplified model

Assume one physical vehicle of each type, a single depot, known static orders and repeated sequential trips. Each customer is served once without splitting its order. Every trip returns to depot and must respect the assigned vehicle's weight and aggregate-volume limits. Vehicle counts, objective and time horizon still require agreement.

The generator simulates repeated transports; the initial optimizer receives the generated orders as one static batch available at time zero. It does not yet model dynamic release or a maximum working day. Schedules use an explicit assumed average speed of 35 km/h and generated service times.

Objective choice matters

The prototype supports total distance and estimated fuel. The default proposal is **total estimated fuel litres**: sum over trips of distance in km × vehicle fuel rate / 100. Rates are sampled once per vehicle and held fixed during search. This is a constant-rate proxy, not a physical fuel-consumption model.

With unlimited trips and no deadline, the largest vehicle can reproduce a smaller vehicle's route at the same distance, so distance alone provides no advantage for using smaller vehicles. Fuel, operating cost, deadlines or availability must make that choice meaningful. The official ranking must be agreed before the final comparison.

2. Representation and algorithm design

Two-part chromosome: order and allocation

Use a customer permutation p and a vector a indexed by customer ID. Changing the position of a customer in p does not change which entry of a belongs to that customer.

Component	Example	Meaning
Visit permutation p	[3, 1, 4, 2]	Each customer appears exactly once.
Allocation a	[0, 1, 0, 2]	Customers 1..4 use vehicles 0, 1, 0 and 2.
Vehicle subsequences	0: [3, 1]; 1: [2]; 2: [4]	Filter p by customer assignment, preserving within-vehicle order.
Decoded solution	Vehicle ID + list of depot tours	Split each subsequence into capacity-feasible trips, then schedule them sequentially.

This explicitly represents both order and vehicle allocation. A single giant tour with a joint fleet-assignment decoder is a literature-backed alternative, but requires more elaborate decoding. Prins [R1] motivates that alternative; the two-vector design is our simpler first implementation.

Implemented operators

Component	Choice
Initialization	Known feasible generated transport witness plus random compatible allocations and orders.
Selection and recombination	Tournament size 3; OX order crossover with probability 0.9; uniform customer-indexed allocation crossover.
Mutation	Swap/reversal of order with probability 0.3; compatible-vehicle reassignment with probability 0.3.
Decoding and education	Per-vehicle dynamic-programming Split; one improving 2-opt reversal per decoded trip.
Survival	Two elite individuals; minimize the selected distance or estimated-fuel objective.

Exactness and complexity boundary

Split is exact only for contiguous trip boundaries of a fixed order and allocation under the implemented additive capacities and unrestricted horizon. For a fixed vehicle rate, minimizing its distance also minimizes its fuel estimate. The joint order/allocation search remains heuristic.

With n customers, fixed fleet size, population P and G generations, the straightforward decoder and bounded local search cost $O(n^2)$ per candidate, giving $O(GPn^2 + GP \log P)$ search time. Working storage includes the $O(n^2)$ matrix and $O(Pn)$ population. Time windows, deadlines or dynamic release would require different state and feasibility logic.

3. Catalog audit and transport generator

Amazon furniture sample: semantic checks before use

The candidate Kaggle dataset was downloaded, de-duplicated and audited. Its public metadata reports CC0: Public Domain. The raw sample, metadata snapshot, raw strings, source rows, checksum and exclusion reasons are retained. [D1]

Audit result	Count / interpretation
Downloaded rows / unique ASINs	312 rows / 305 unique ASINs.
Repeated or empty-ASIN rows	7 rows.
Proxy-capable furniture records	193 records with parseable three-axis dimensions and item/shipping weight.
Strict verified-package rule	0 records with both explicit Package Dimensions and shipping/package weight.
Runnable initial data	Explicit --allow-proxy mode, with limitations propagated to generated instances.

The column named package_dimensions can repeat Product Dimensions; item weight is not necessarily shipping weight. Product sets and multiple cartons also need interpretation. The audit does not prove that the provider's entire dataset has no suitable packaging; it describes this sample and the stated strict rule.

Generation procedure

For each physical vehicle: sample a target stop count, select appropriately small furniture units, fill a load while respecting both capacities, divide it into customer orders, generate locations and service times, visit stops in a shuffled order, return to the depot, and begin the next load. Each generated tour is retained as a feasible aggregate-capacity witness.

Generated quantity	Normal model / handling
Stops per transport	Mean 10, SD 2; bounded 3..20 and rounded to an integer.
Limiting-resource utilization	Mean 0.8, SD 0.07; bounded 0.6..0.95.
Synthetic item dimensions / weight	Catalog means; CV 3% / 5%; positive bounded draws.
Coordinates / service time	Coordinates: mean 0, SD 10 km, bounded ±30 km. Service: mean 10, SD 2 minutes, bounded 2..30.
Vehicle fuel rate	Normal around supplied range midpoint, SD equal to range width / 6, bounded by that range.

Included development case: seed 42, two loads per vehicle, six transports, 64 customers and 437 generated furniture units. Stops are 11, 10, 12, 11, 12 and 8: mean 10.67. A separate seed-43 validation input is also supplied. Truncation, catalog filtering and capacity rejection condition the distributions; generated loads are not unconditioned Gaussian observations.

4. Research supporting the choices

Reference	Research contribution	Connection to this implementation
[R1] Prins (2009)	Memetic methods for heterogeneous fleet routing, with giant-tour splitting and vehicle assignment.	Supports studying the interaction between order, trip boundaries and vehicle choice.
[R2] Vidal (2022)	Hybrid genetic search for CVRP and SWAP* neighborhood.	Supports population search plus route improvement; this prototype does not reproduce full HGS.
[R3] Zhao et al. (2024)	Genetic search and dynamic-programming Split for multi-trip time-dependent routing.	Motivates multi-trip decoding; our initial case omits time dependence and deadlines.
[R4] Penna et al. (2018 preprint)	Hybrid ILS, variable-neighborhood descent and set partitioning for heterogeneous rich VRPs.	A relevant alternative algorithm family, not implemented here.
[R5] Zhao, Bian and Mei (2024)	Adaptive multi-objective GA for heterogeneous green city routing.	Useful targeted reading for operating/environmental objectives; our fuel proxy is simpler.
[R6] Hvattum (2025)	Effects of giant-tour starting points on HGS splitting.	A focused recent refinement to investigate after the basic representation works.

Algorithm-family understanding

Genetic and memetic approaches evolve populations through crossover, mutation and local improvement. ILS/VNS repeatedly perturb and improve route plans. ALNS destroys and repairs selected parts of a solution. Swarm approaches such as ACO or discrete PSO require discrete constructions or encodings that satisfy the same routing model. These families should be compared through a common evaluator, not differently simplified problems.

What has and has not been implemented

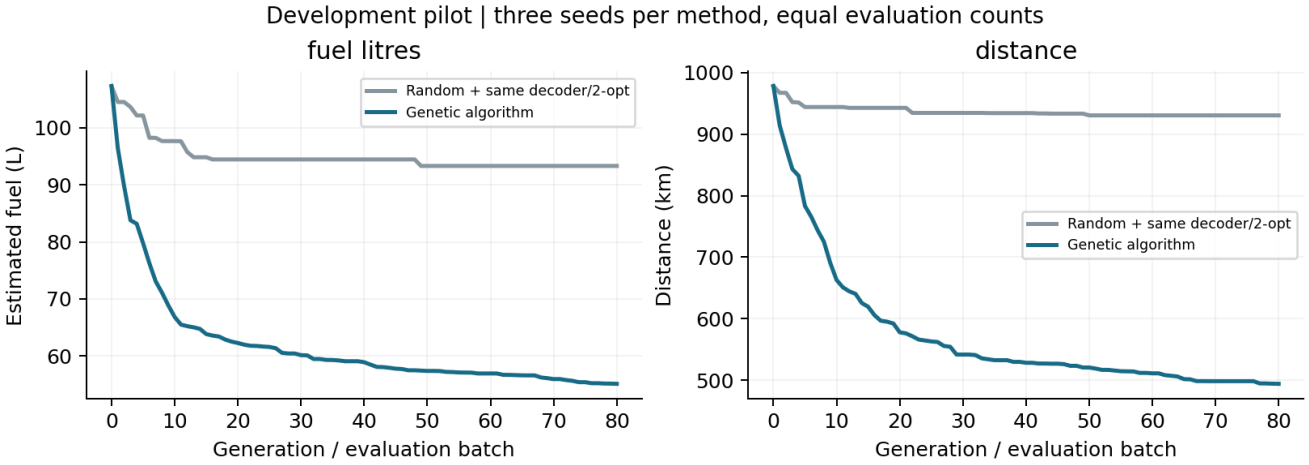
The supplied initial algorithm is a small, explicit two-part genetic search with capacity-feasible decoding and sequential trip schedules. It includes a random-search baseline with the same decoder and bounded 2-opt. It does not implement a swarm algorithm, an exact global optimizer, geometric packing, real road travel, dynamic orders or driver-shift limits.

There is no certified optimum for the newly generated case. Report distances, estimated fuel, feasibility, runtime and evaluation counts. A baseline reduction is not an optimality gap. The earlier 96.100 km homogeneous demonstration is a different input/model and cannot be reused as a reference objective.

5. Initial implementation and observed pilot

The development pilot uses one fixed 64-customer input, two objectives, two search methods and three seeds: 12 runs in total. Each run uses population 32, 80 generations and 2,432 candidate evaluations. GA and random search share initialization rules, decoder and bounded 2-opt. This controls evaluation counts; it is not an equal-runtime or held-out final comparison.

Objective	Search	Mean km	Mean fuel L	Mean s
Fuel	Random	959.663	93.322	0.703
Fuel	GA	616.932	55.118	0.538
Distance	Random	930.649	117.186	0.717
Distance	GA	493.985	64.526	0.732



Selected fuel-objective solution

The best fuel-objective GA run (seed 2) uses all 3 vehicles across 13 sequential trips. Its distance is **619.695 km** and estimated fuel is **51.262 L**. Every customer is served exactly once and all per-trip weight/volume constraints pass independent validation.

Four generator tests and six algorithm tests passed. They cover reproducibility, unit parsing, explicit proxy opt-in, generated capacity witnesses, customer-indexed allocation, exhaustive small fixed-order Split validation, coverage/overload rejection and nonoverlapping trips.

Schedules can extend beyond a working day because no daily limit was supplied or implemented. These results demonstrate runnable initial code under stated assumptions, not operationally validated fuel savings or superiority over the swarm team. Test unseen instances and agree budgets before ranking algorithms.

6. Deliverables, execution and references

Repository area	Included work
transport-findings	Problem definition, representation rationale, algorithm-family review, references and meeting script.
transport-generator	Raw sample and audit, vehicle inputs, normal-based generator, development/validation cases and four tests.
transport-algorithms	Two-part GA, decoder, validator, six tests, 12-run pilot, solution JSON and optional figures.

Run the first implementation

From transport-generator:
python generate.py --catalog data/catalog_proxy.json --allow-proxy --cycles 2 --seed 42 --output data/development.json

From transport-algorithms:
python solver.py
python -m unittest -v test_solver
python run_experiments.py

Python 3.10+ and the standard library are sufficient for generation and solving. Matplotlib is optional for plots. The algorithm repository carries a frozen copy of the generator input so it runs independently.

GitHub publication status

The three areas are prepared as independent local Git repositories with portable bundles. **No remote GitHub repository has been created or pushed.** The available connector does not expose repository creation. The package includes instructions for creating three empty destinations and pushing the completed contents. Publication remains the final pending item.

Primary references

[R1] Prins, C. (2009). Two memetic algorithms for heterogeneous fleet vehicle routing problems. Engineering Applications of Artificial Intelligence, 22(6), 916-928. https://doi.org/10.1016/j.engappai.2008.10.006

[R2] Vidal, T. (2022). Hybrid genetic search for the CVRP: Open-source implementation and SWAP* neighborhood. Computers & Operations Research, 140, 105643. https://arxiv.org/abs/2012.10384

[R3] Zhao, J., Poon, M., Tan, V. Y. F., and Zhang, Z. (2024). A hybrid genetic search and dynamic programming-based split algorithm for the multi-trip time-dependent vehicle routing problem. EJOR, 317(3), 921-935. https://doi.org/10.1016/j.ejor.2024.04.011

[R4] Penna, P. H. V., et al. (2018 preprint). A Hybrid Heuristic for a Broad Class of Vehicle Routing Problems with Heterogeneous Fleet. https://arxiv.org/abs/1803.01930

[R5] Zhao, W., Bian, X., and Mei, X. (2024). An Adaptive Multi-Objective Genetic Algorithm for Solving Heterogeneous Green City Vehicle Routing Problem. Applied Sciences, 14(15), 6594. https://doi.org/10.3390/app14156594

[R6] Hvattum, L. M. (2025). Where to Split in Hybrid Genetic Search for the Capacitated Vehicle Routing Problem. Algorithms, 18(3), 165. https://doi.org/10.3390/a18030165

[D1] Crawl Feeds. Furniture products dataset from amazon (Kaggle). https://www.kaggle.com/datasets/crawlfeeds/furniture-products-dataset-from-amazon